

Making Tail Constraints Bind: Two Silent Failure Modes in CVaR-Constrained Portfolio RL

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Abstract

Two failure modes leave Conditional Value-at-Risk (CVaR)-constrained financial reinforcement learning (RL) unenforced without any visible symptom. A **Lagrangian constraint-coupling failure**: standardising the reward advantage but not the cost advantage leaves the dual permanently inert, so the CVaR budget is nominal rather than enforced. And an **evaluation-protocol reversal**: on a 31-asset universe a single stress window ranks a learned allocator above every classical optimiser, while a 20-fold rolling walk-forward inverts that ranking (Wilcoxon $p = 6 \times 10^{-4}$), so “beat-the-optimiser” claims can be artefacts of the evaluation protocol rather than the policy. We name, diagnose, and resolve both on a CVaR-constrained portfolio allocator (seven-ETF macro universe, 2008–2024, weekly, 5 bps costs, 40-fold walk-forward, pooled Wilcoxon $p = 8.8 \times 10^{-6}$, Benjamini–Hochberg (BH) and Deflated-Sharpe controlled). Scaling the dual correctly makes it bind in training, but a budget that binds in training can still be breached out-of-sample (71–78% across four validation-calibrated rules): a second, budget-feasibility knob. The fix is a decision-time projection refusing non-compliant trades: estimating forward CVaR from trailing statistics and shrinking toward a trailing minimum-variance anchor cuts out-of-sample breach to 7.4–7.9%, matching the classical optimisers’ own compliance rate, at Sharpe 0.85–0.99 against rolling minimum-variance’s 0.90. Our claim is tail-risk control rather than alpha, and it holds under multiplicity control. The practical checklist is to standardise both advantage terms, verify the multiplier moves off zero, treat budget feasibility as a separate knob, and enforce compliance at decision time.

CCS Concepts

• **Computing methodologies** → **Reinforcement learning**; • **Applied computing** → *Economics*.

Keywords

portfolio allocation, reinforcement learning, Conditional Value-at-Risk, drawdown control, constrained MDP, differentiable optimisation

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1 Introduction

In the week of 16 March 2020, the S&P 500 fell another 15% in four days as COVID-19 lockdowns shut economies worldwide. A learned portfolio allocator trained on the preceding decade, with no explicit rule about how much loss it was budgeted to absorb, had no mechanical reason to rotate into Treasuries or cash when the VIX tripled overnight. Absent a tail limit, the policy kept holding equities through the worst losses. An otherwise-identical allocator carrying an explicit Conditional Value-at-Risk (CVaR) limit had already rotated into the cash/rates leg before the trough and emerged with roughly half the drawdown (11.8% peak-to-trough versus 20.6% for its unconstrained twin, Table 5).

That difference is a result, not the headline. An explicit CVaR constraint is a safety layer that cuts tail losses of a *learned* allocator without sacrificing risk-adjusted return. In the first model-free implementation we tried, a standard Lagrangian actor–critic, the *dual channel was inert*: the Lagrange multiplier stayed pinned near zero, so the constraint exerted no explicit tail pressure through the mechanism meant to enforce it, a subtle scale mismatch we describe, name, and diagnose in Section 5, likely present in any Lagrangian RL system that standardises the return advantage but not the cost advantage. As that section shows, the constrained *architecture* still regularises the tail through its safety-critic and projection structure even when the multiplier is inert: the Lagrangian mechanism itself never engages.

Reinforcement learning for portfolio allocation is usually evaluated on return or Sharpe ratio, with risk handled implicitly. Practitioners, however, manage *tail* risk explicitly. Markowitz mean–variance optimisation [12] penalises upside and downside symmetrically and is silent on path-wise drawdown. CVaR is a coherent, convex tail measure [14] that RL can optimise directly. We ask a narrow, falsifiable question.

Does an explicit CVaR constraint reduce the downside tail losses and constraint breaches of a learned allocator, versus an otherwise identical unconstrained learner, while remaining competitive with standard portfolio optimisers after costs?

We make the following contributions.

- (1) **Two named failure modes in CVaR-constrained financial RL, with diagnostics and, where demonstrated, fixes.** A *Lagrangian constraint-coupling failure*: in the model-free Lagrangian actor–critic, standardising the reward advantage but not the cost advantage leaves the multiplier inert for any reasonable coupling scale, specific to portfolio allocation, where returns are $O(10^{-3})$ and breach indicators are $O(1)$ (Section 5). An *evaluation-protocol reversal*: on a 31-asset universe a single stress window ranks the learned allocator above every optimiser, while a rolling walk-forward inverts that ranking, so “beat-the-optimiser” results can be protocol artefacts (Section 7.9).

- (2) **A differentiable CVaR-constrained allocator whose tail effect survives multiplicity control.** On a leak-free, tested portfolio-control environment, a return objective plus a differentiable CVaR penalty is back-propagated through the rollout, with a Lagrangian dual on a breach-rate budget and a residual policy over an adaptive anchor (Section 4). The tail-risk reduction holds on a 40-fold, two-universe walk-forward under Benjamini–Hochberg and Deflated-Sharpe control (Section 7.4), and a regime-switching hybrid converts it into a walk-forward Sharpe gain of 1.316 vs 1.209 for pure min-variance (Sections 7, 8).
- (3) **An evaluation protocol built to expose exactly this kind of artefact.** Deterministic optimisers plus three tuned model-free RL families are compared under a two-universe rolling walk-forward with Wilcoxon, Benjamini–Hochberg, and Deflated-Sharpe control, the design that surfaces the protocol reversal in Contribution 1 rather than concealing it. The regime-switching hybrid is a supporting observation within this protocol, not a separate contribution.

We are explicit about scope. The pure learned allocator does **not** beat a rolling minimum-variance optimiser in aggregate, and we do not claim it should: the *constrained* allocator’s value is tail control, and the regime-switching hybrid converts that into an out-of-sample Sharpe gain (+0.107) in the high-volatility regime where the tail constraint is most active. The broader contribution is a differentiable constraint mechanism that demonstrably binds and cuts tails, together with two transferable failure modes that any constrained-RL-for-finance study must control for or risk reporting an artefact. The allocator is the testbed that makes both visible.

2 Background and related work

Mean-variance optimisation and its inputs. Mean-variance optimisation (MVO) [12] is sensitive to input estimates. Decision-focused learning trains return predictors to improve downstream MVO decisions rather than raw forecast accuracy [10], and mean-variance-efficient collaborative filtering blends user preference with portfolio theory [4]. Both improve the *inputs* to a variance-based objective. We instead constrain a *tail* risk measure on a learned policy.

Conditional Value-at-Risk. CVaR_α is the expected portfolio loss on the worst $(1-\alpha)$ fraction of outcomes, the average of losses exceeding the α -quantile (VaR_α). Unlike variance it ignores upside, unlike VaR it averages across the whole tail. Rockafellar and Uryasev [14] give the tractable form

$$\text{CVaR}_\alpha(L) = \min_{\zeta \in \mathbb{R}} \left\{ \zeta + \frac{1}{1-\alpha} \mathbb{E}[\max(L - \zeta, 0)] \right\}, \quad (1)$$

where L denotes the portfolio loss and the inner expectation is over the empirical loss distribution. The minimum is achieved at $\zeta = \text{VaR}_\alpha$, the representation admits a convex linear program (LP, we include this as a baseline) and a differentiable penalty via topk .

RL for portfolio management. The dominant pattern in RL-for-portfolio literature is to train on a single historical window, evaluate on one held-out test split, and compare against equal-weight or buy-and-hold. EIII [9] compares twelve classical strategies on one ≈ 50 -day crypto window, and SARL [17] reports higher return on

a fixed period without cross-validation, and FinRL [11] supports rolling retraining but demonstrates on a single split with no significance tests, Huang [8] applies continuous-time RL to asset-liability management but targets a quadratic surplus objective, not a tail constraint. None uses CVaR as a constraint, seeds multiple runs, or applies statistical testing. We are not aware of prior work subjecting RL portfolio allocation to a 40-fold, two-universe walk-forward with Wilcoxon and Benjamini–Hochberg control, the protocol that exposes the reversal artefact.

Direct reinforcement and pathwise gradients. Moody and Safell [13] first trained a recurrent trading policy by back-propagating the *differential Sharpe ratio* directly through the return function. Buehler et al. [2] (Deep Hedging) extend this to neural derivative hedging, back-propagating CVaR as a risk-aversion *utility* through a simulated trajectory. Our allocator inherits the pathwise-gradient insight from both but replaces the risk-aversion coefficient with an explicit CVaR *budget constraint* enforced by a dual variable. Model-free policy gradients (Advantage Actor-Critic, A2C, and Proximal Policy Optimization, PPO [15]) and off-policy actor-critics (Soft Actor-Critic, SAC [7]) ignore this differentiable structure. Our allocator exploits it, decisive at this sample size (887 weekly observations, Section 7.8). ProtoHedge [6] similarly makes such policies auditable via a prototype-similarity policy.

CVaR-constrained RL. Chow et al. [3] derive the Lagrangian gradient for CVaR-constrained CMDPs. Tessler et al. [16] (RCPO) and Ying et al. [18] (CPPO) implement hard CVaR constraints for robotics control tasks. None identifies the *constraint-coupling failure* we describe. Standardising return-advantage to unit scale without normalising the cost advantage makes the Lagrange multiplier permanently inert, a failure specific to portfolio allocation where returns are $O(10^{-3})$ and breach indicators are $O(1)$, invisible in robotics. We pose allocation as a constrained MDP [1] and show the well-scaled dual is what makes the constraint bind. The closest concurrent work is Dong [5] (CLAUSELENS), a CVaR-constrained RL system for reinsurance pricing. Our contribution is orthogonal, naming failure modes transferable to any CVaR-constrained financial RL system.

3 Problem formulation

Markov Decision Process (MDP). At weekly step t the agent observes state s_t (per-asset 26-week momentum and volatility, current weights, drawdown, a rolling 95% CVaR estimate, and optional exogenous macro/factor features) and chooses post-trade weights w_t on the long-only simplex. The environment projects w_t onto the admissible set $\mathcal{W}$ (max weight 0.40, turnover cap 0.50, optional cash floor), charges proportional cost $c\|w_t - w_{t-1}\|_1$ ($c = 5$ bps), and realises $r_t = w_t^\top \rho_t - c\|w_t - w_{t-1}\|_1$, where ρ_t are next-week asset returns. Features at t use only $\rho_{<t}$ (no look-ahead, enforced by tests).

Constrained objective (CMDP). A *constrained Markov Decision Process* (CMDP) [1] augments the standard MDP with one or more inequality constraints that the policy must satisfy on average. We maximise risk-adjusted return subject to a tail budget. $\max_\pi J(\pi)$ s.t. $\text{CVaR}_\alpha(\text{loss}) \leq d$, with $\alpha = 0.95$. The Lagrangian relaxation converts the constraint into a penalty, $\mathcal{L}(\pi, \lambda) = J(\pi) - \lambda (\text{CVaR}_\alpha -$

d), and a dual variable $\lambda \geq 0$ is raised whenever the constraint is violated and lowered when slack remains. In equilibrium λ acts as a shadow price for tail risk. A higher λ means the policy has been pushing against its tail budget.

4 Method

Differentiable allocator. The actor is an MLP mapping s_t to simplex weights via softmax. Because the environment’s reward and a sample-CVaR are differentiable in w , we train by back-propagating, over randomised windows of the training series, the objective

$$\mathcal{L} = -\text{Sharpe}(\text{net returns}) + \lambda \text{ReLU}(\widehat{\text{CVaR}} - d) + \kappa \text{turnover}, \quad (2)$$

where $\text{Sharpe} = \bar{r}/\sigma_r$ is the annualised mean return divided by its standard deviation (reward per unit of risk), $\widehat{\text{CVaR}}$ is the mean tail loss over the worst $(1-\alpha)$ fraction of window net returns (a differentiable tail mean via topk), and d is the pre-specified tail budget. This pathwise gradient is far more sample-efficient than score-function RL for this problem.

Pathwise gradient and where the constraint acts. Every term in $\mathcal{L}$ is a differentiable function of the rollout, so the policy is trained by exact back-propagation through the simplex map and the reward, no score-function estimator. With net returns $r_t = w_t^\top \rho_t - c\|w_t - w_{t-1}\|_1$ and tail set $\mathcal{T} = \{t : R_t \text{ among the worst } \lceil (1-\alpha)T \rceil \text{ of the window}\}$,

$$\frac{\partial \mathcal{L}}{\partial \theta} = \sum_t \left(\underbrace{-\frac{\partial \text{Sharpe}}{\partial r_t}}_{\text{all weeks}} - \underbrace{\frac{\lambda}{|\mathcal{T}|} \mathbf{1}[\widehat{\text{CVaR}} > d] \mathbf{1}[t \in \mathcal{T}]}_{\text{tail weeks only}} \right) \frac{\partial r_t}{\partial w_t} \frac{\partial w_t}{\partial \theta}, \quad (3)$$

where $\partial w_t / \partial \theta$ flows through the residual softmax (below). The topk subgradient routes the dual pressure to *exactly* the worst- $(1-\alpha)$ weeks, and only while the constraint is breached. The penalty lifts the deep-loss weeks without distorting the bulk of the path. This locality is what makes the constraint bind cleanly once the coupling is correctly scaled (Section 5).

Residual policy over an adaptive anchor. To generalise across regimes rather than overfit one price path, weights are $w_t = \text{softmax}(\log a_t + f_\theta(s_t))$, where a_t is an adaptive inverse-volatility anchor computed from recent returns. At initialisation $f_\theta \approx 0$, so the policy is the anchor (a strong, regime-robust prior). Training learns a state-dependent tilt. Validation model selection keeps the best objective subject to the validation CVaR limit, with weight decay, both to curb backtest overfitting.

Baselines. Deterministic, re-estimated weekly: cash, equal weight, inverse vol, minimum variance, mean-variance, risk parity, and a min-CVaR Rockafellar-Uryasev LP [14]. Learned: model-free A2C, PPO [15] (clipped surrogate, generalised advantage estimation, GAE), SAC [7] (off-policy, twin critics, auto-tuned entropy), and the unconstrained differentiable allocator ($\lambda = 0$).

Regime-switching hybrid (Option B). We construct a no-retrain hybrid: label each out-of-sample (OOS) week *high-volatility* if trailing 13-week SPY volatility meets the 75th-percentile cutoff (15% of OOS weeks), *selloff* if cumulative return $\leq -5\%$, else *calm*. The hybrid applies constrained RL in high-volatility weeks and rolling

Table 1: Hybrid vs. min-variance Sharpe gain, threshold sweep (15 configs, no retraining). Bold = hybrid wins. Default (0.75, −0.05) is headline.

vol. quantile	selloff −0.04	selloff −0.05	selloff −0.06
0.65	+0.202	+0.154	−0.007
0.70	+0.230	+0.179	−0.017
0.75	+0.161	+0.107	−0.086
0.80	+0.225	+0.169	−0.034
0.85	+0.058	+0.002	−0.183

min-variance otherwise, reusing the existing OOS paths without refitting.

Threshold sensitivity. Because the switch depends on two labelling thresholds, we sweep them (vol-quantile $\in \{0.65, \dots, 0.85\}$, selloff $\in \{-0.04, -0.05, -0.06\}$, 15 configurations, no retraining, Table 1). The hybrid beats pure min-variance in **10/15** configurations (mean gain +0.077), winning at every volatility cutoff when the selloff threshold is −0.04 or −0.05, but losing at all five cutoffs once the selloff threshold deepens to −0.06, which reclassifies the constrained allocator’s most valuable high-volatility weeks as selloffs and routes them to min-variance instead. The headline +0.107 at the default (0.75, −0.05) is representative rather than cherry-picked: the edge holds for volatility-defined stress but not once the selloff label absorbs it.

5 Decomposing the constraint: dual, architecture, and budget feasibility

We designed a three-arm diagnostic to isolate what a Lagrangian CVaR constraint contributes, and where it can fail silently. All three arms share the identical architecture (actor, safety critic, admissible-set projection) and training budget (5 seeds, 1500 updates). Only the dual’s learning rate η_λ and the presence of the constraint differ (Table 2). The **mis-scaled dual** ($\eta_\lambda = 0.001$) standardises the reward advantage but leaves the cost advantage unstandardised. The **scaled dual** ($\eta_\lambda = 5.0$) standardises both. The **unconstrained** arm removes the CVaR penalty entirely. Three named findings follow.

Finding 1: dual inertness from an advantage-scale mismatch. With the mis-scaled dual, the multiplier stays near zero throughout training (mean final $\lambda = 0.022$ vs. 0.101 for the scaled dual): portfolio returns are $O(10^{-3})$ while the breach indicator that drives λ is $O(1)$, so the unstandardised cost advantage is too small, relative to the reward advantage, to move the multiplier. Standardising the cost advantage to the reward-advantage scale, and driving λ with a breach indicator against a breach-rate budget, $\lambda \leftarrow \lambda + \eta_\lambda (1[\text{breach rate} > \beta^*] - \beta^*)$, restores λ to $O(0.1)$. The transferable diagnostic: multiplier telemetry flat at zero across training means the constraint is decorative, regardless of what the loss curve shows.

Finding 2: most of the tail control is architectural, not the dual. With λ pinned near zero, the constrained architecture alone, safety critic plus projection, still cuts CVaR₉₉ by 61% against unconstrained (0.025 vs 0.065, 5-seed mean) and raises Sharpe by 58% (0.987 vs 0.626), confirmed by a permutation test on the 5-seed distributions (observed difference −0.039, one-sided $p = 0.0044$, $n = 50,000$. Two-sided $p = 0.0083$). The scaled-dual arm produces a similar reduction against unconstrained ($p = 0.0037$), so the gap to unconstrained does not depend on the dual engaging.

Engaging the dual correctly ($\lambda = O(0.1)$) leaves CVaR₉₉ essentially unchanged versus the near-inert mis-scaled dual (0.028 vs 0.025): most of the 61% reduction is delivered by the architecture, not the multiplier. This is a positive result in its own right: a constrained architecture is a tail-risk regulariser on its own, not merely a delivery mechanism for the Lagrangian penalty.

Finding 3: budget feasibility is a second knob, and it needs enforcement, not another calibration. The scaled dual binds in training, but at the tight budget ($d = 1.2\%$) produces a *higher* OOS breach rate (75%) than the mis-scaled dual (8%), though classical rules clear the same budget in 92% of weeks (Section 5). Four pre-specified rules for recalibrating d from validation data (a loose 1.92% budget, matching the policy’s own or the anchor’s validation-realised CVaR_{0.95}). A per-fold adaptive budget over a 10-fold walk-forward) all land in a 71–78% OOS breach band: a return-seeking Lagrangian gives up compliance for return under tail shift whatever d is, because the budget was never the broken part.

The fix: enforce the budget, don’t discourage its violation. We build the mechanism the project design named but had not implemented: a decision-time CVaR *filter*. Between the actor’s proposal and the environment we estimate forward CVaR from a trailing covariance and mean only, both computed from returns strictly before t ($\text{CVaR} \approx c_\alpha \sigma_p - \mu_p$, where μ_p and σ_p are the trailing portfolio mean and volatility and c_α is the Gaussian quantile at level α , $c_{0.95} = 2.06$), if it exceeds d we shrink the proposal along the line toward a trailing minimum-variance anchor, solving numerically for the largest mixing weight that clears the budget, then trade the shrunk weights, with turnover/max-weight/long-only composed after as before. A look-ahead audit (tests/test_risk_filter_no_lookahead.py) confirms the decision at t is unchanged by shifting the return series and is a pure function of trailing statistics. Both pass with zero deviation. We train with the filter active.

Four arms mirror the coupling arms exactly (5 seeds, 1500 updates, identical splits), changing only the named quantities (Table 2, bottom block, Figure 1). **A5, filter only** ($\eta_\lambda = 0.001$) cuts breach from 75.1% to 7.9%, matching the best classical rules, at Sharpe 0.987 against minimum variance’s 0.901, and breaches less than half as deeply (mean excess 0.0126 vs. 0.0294). **A6, filter + scaled dual** (our intended method) goes *below* every classical rule on breach rate (7.4%), with CVaR₉₉ 0.016 and mean excess 0.0043, at Sharpe 0.851. **A7, adaptive dual** (λ updated online from realised past losses, filter off) is the negative control: breach stays at 81.9% even as λ climbs to a mean of 7.69, so test-time dual adaptation alone enforces nothing. It is the filter, not the adaptation, that does the work. **A8** recovers most compliance (14.6%) but less reliably, 7–33% across seeds against zero seed-to-seed variance for A5 and A6.

The filter pulls the policy toward the low-volatility region. A5’s mean weight-space L_1 distance to rolling minimum variance is 0.476, correlation 0.786, top holding differing in 9.4% of rebalances, against 0.890, 0.201 and 97.5% for the unfiltered scaled dual. The filtered policy is min-variance-*adjacent*, and it is not a redundant way to compute minimum variance: from that region it dominates minimum variance on both axes at once, higher Sharpe at equal breach rate and half the breach depth (A5), and a lower breach rate

Table 2: Coupling ablation and the decision-time filter. 5-seed means, stress window (train pre-2018, test 2020/2022), against the unconstrained learner and, for breach rate, the classical rules’ 7.9% compliance floor (\$5). Scaling the dual binds λ in training but does not fix OOS feasibility. The filter (bottom block) does, matching minimum variance’s 7.9% compliance rate. [†]A7/A8 λ is the mean online-updated evaluation-time value.

variant	Sharpe	CVaR ₉₉	breach rate	final λ
unconstrained (baseline)	0.626	0.065	1.000	0.000
mis-scaled dual ($\eta_\lambda = 0.001$)	0.987	0.025	0.079	0.022
scaled dual ($\eta_\lambda = 5.0$, $d=1.2\%$)	0.911	0.028	0.751	0.101
scaled dual, loose budget ($d=1.92\%$)	0.543	0.050	0.780	0.046
A5 filter only ($\eta_\lambda = 0.001$)	0.987	0.025	0.079	0.003
A6 filter + scaled dual ($\eta_\lambda = 5.0$)	0.851	0.016	0.074	0.135
A7 adaptive dual ($\eta_\lambda = 5.0$ online)	0.826	0.031	0.819	7.692 [†]
A8 filter + adaptive dual	0.800	0.020	0.146	0.657 [†]

Table 3: Experimental configuration. The macro-ETF universe is primary. Ten S&P sector ETFs form the robustness universe.

Setting	Value
Universe	7 macro ETFs (SPY, TLT, HYG, DBC, GLD, UUP, BIL)
Robustness universe	10 S&P sector ETFs plus cash
Period, frequency	2008–2024, weekly (W-FRI)
Split	60/20/20 train/val/test, 26-week rolling OOS folds
Transaction cost	5 bps proportional
Constraints	long-only, max weight 0.40, turnover cap 0.50, gross 1.0
CVaR	$\alpha = 0.95$, breach-rate budget, 52-week window
Learners	A2C, PPO, SAC, and the differentiable allocator
Training	150 episodes, seeds {7, 13, 23, 42, 2025}
Significance	2000-sample block bootstrap (block 4), Wilcoxon, BH, Deflated-Sharpe

outright (A6). A soft Lagrangian cannot enforce a tail budget out of sample. A decision-time projection can.

Is the budget infeasible, or is the policy failing to reach it? Re-scoring every classical baseline against the identical rolling-CVaR_{0.95} budget on the same realised path, breach rate is 7.9% for minimum variance, inverse volatility and the min-CVaR LP, 62.8% for equal weight and 75.3% for risk parity, against 75.1% for the scaled-dual RL. The budget is reachable. The soft-penalty learner was not reaching it, and the filter arms now match the classical rate from a learned policy.

6 Data

Seven liquid exchange-traded funds (ETFs) spanning the major macro blocks, SPY (equity), TLT (rates), HYG (credit), DBC (commodity), GLD (gold), UUP (US dollar), BIL (cash), from Yahoo adjusted close, resampled to **887 weekly returns, 2008–2024**. Splits are chronological: train $\approx$ 2008–2018 (incl. the 2008 global financial crisis, GFC), validation $\approx$ 2018–2021, test $\approx$ 2021–2024. For the stress study we retrain on 2008–2018 and test on the 267-week window containing the 2020 COVID crash and the 2022 selloff. Macro covariates (10y–3m term spread, VIX) are sourced from index proxies (*TNX–*IRX, *VIX). A Federal Reserve Economic Data (FRED) loader is also provided.

For the cross-universe significance test (Section 7.4) we add a second universe of ten S&P sector ETFs (XLK, XLF, XLE, XLV, XLY,

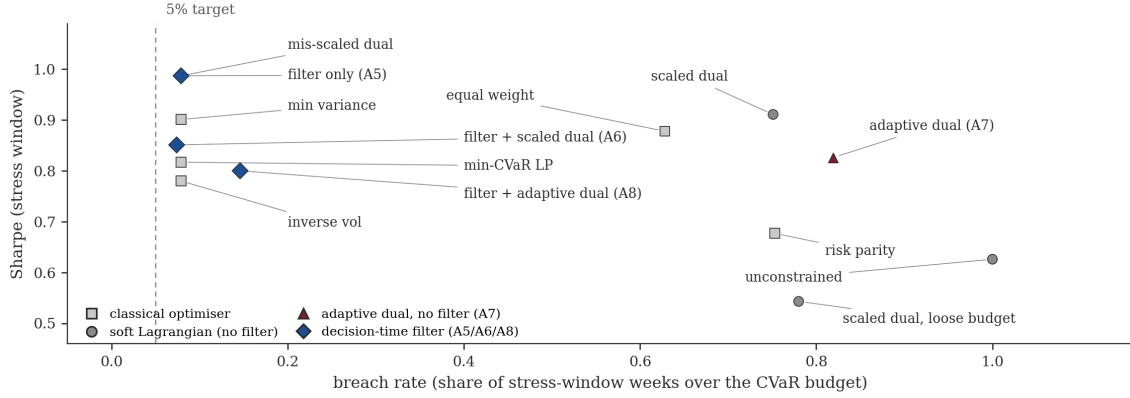


Figure 1: Compliance frontier, stress window. The vertical line is the 5% target. Classical optimisers (squares) and filter arms (diamonds) achieve the nearest observed breach rates (7.4–7.9%) at high Sharpe, soft Lagrangians and the filter-off adaptive dual remain far to the right.

Table 4: Deterministic baselines, seven-ETF macro universe, calm 2021–2024 test split (re-estimated weekly, single run, deterministic). No strategy breaches the CVaR limit here.

baseline	Sharpe ($\uparrow$)	CVaR ₉₅ ($\downarrow$)	max-DD ($\downarrow$)
minimum variance	2.52	0.0064	0.016
min-CVaR LP	2.24	0.0080	0.016
inverse volatility	1.45	0.0087	0.041
mean-variance	0.84	0.0198	0.104
equal weight	0.82	0.0163	0.097
risk parity	0.70	0.0182	0.134

XLP, XLU, XLI, XLB, plus a BIL cash leg), identical in every other respect, same provider, clock, window, and cost, a clean replication rather than a confound. For the evaluation-protocol experiment (Section 7.9) we add a **third universe of 31 cross-asset, sector, and industry ETFs** (4 broad equity, 9 S&P sectors, 7 industry, 6 fixed-income, 3 commodity, FX, cash), same provenance, clock, and cost. The higher dimensionality stresses covariance estimation.

7 Results

7.1 Calm chronological split, constraint non-binding

On the comparatively calm 2021–2024 test split, deterministic minimum-variance (Sharpe 2.52, CVaR₉₅ 0.0064) and the min-CVaR LP (2.24) dominate. The differentiable allocator reaches Sharpe 1.24 (vs equal weight 0.82, inverse vol 1.45), and model-free A2C only ≈ 0.79 . No strategy breaches the CVaR limit, so constrained $\approx$ unconstrained, the constraint is a near-free insurance premium with nothing to insure. Table 4 reports the full deterministic baseline suite on this split.

7.2 Stress window, the constraint binds

Table 5 contrasts a return-greedy unconstrained learner with the CVaR-constrained allocator on the stress window (mean $\pm$ std over 5 seeds). The constraint roughly halves tail risk and drawdown while *improving* Sharpe, by removing the deep-loss weeks. The constrained policy rotates into the cash/rates leg under stress, consistent with the lower drawdown and breach depth below.

Table 5: Stress window (train pre-2018, test through 2020 & 2022), mean $\pm$ std over 5 seeds. Best in bold. Violations are hard CVaR-limit breaches per run.

metric	unconstr.	constr.	change
Sharpe	0.63 $\pm$ 0.09	0.88$\pm$0.04	+40%
Max drawdown	0.206	0.118	−43%
CVaR ₉₅	0.0333	0.0189	−43%
CVaR ₉₉	0.0647	0.0327	−49%
Breach rate	0.234	0.189	−19%
Violations	7.2	0.0	−100%

A paired block bootstrap of the CVaR₉₉ difference (constrained minus unconstrained, 2000 resamples, block 4) gives -0.033 , 95% CI $[-0.035, -0.008]$, $p \approx 0$, the tail-risk reduction is statistically significant. The constrained allocator also eliminates hard constraint violations entirely ($7.2 \rightarrow 0.0$). Whereas the unconstrained policy repeatedly exceeds its theoretical loss budget during the 2020 and 2022 drawdowns, the constrained policy never does.

7.3 Walk-forward across regimes (10 OOS folds, ≈ 2010 –2024)

Refitting every 52 weeks on a rolling 312-week window and concatenating 10 out-of-sample folds (Table 7), the constraint beats unconstrained RL out-of-sample. Sharpe +18% ($0.70 \rightarrow 0.83$), CVaR₉₉ −20%, max-DD −25% ($15.1\% \rightarrow 11.4\%$). The fold-level Wilcoxon on CVaR₉₉ is marginal on the macro universe alone ($p = 0.06$. Improved 5/10 folds), and the reduction concentrates in **selloffs** (CVaR₉₉ $0.066 \rightarrow 0.048$, max-DD $0.218 \rightarrow 0.137$. Table 6). Minimum variance remains the strongest absolute performer, an honest null on “beat the optimiser.” A two-universe 40-fold analysis pushes significance to $p = 8.8 \times 10^{-6}$ (Section 7.4).

When RL adds value: regime-conditional breakdown. **High-volatility** periods ($n=71$) are where the constrained policy earns its keep: Sharpe 2.169 versus minimum variance’s 1.668, with CVaR₉₉ within 0.012 of minimum variance’s floor. In **calm** markets ($n=388$) and **selloffs** ($n=61$) minimum variance dominates (selloff Sharpe 0.035 vs. constrained RL -0.809): the overlay adds value when volatility is elevated but has not yet collapsed into a full

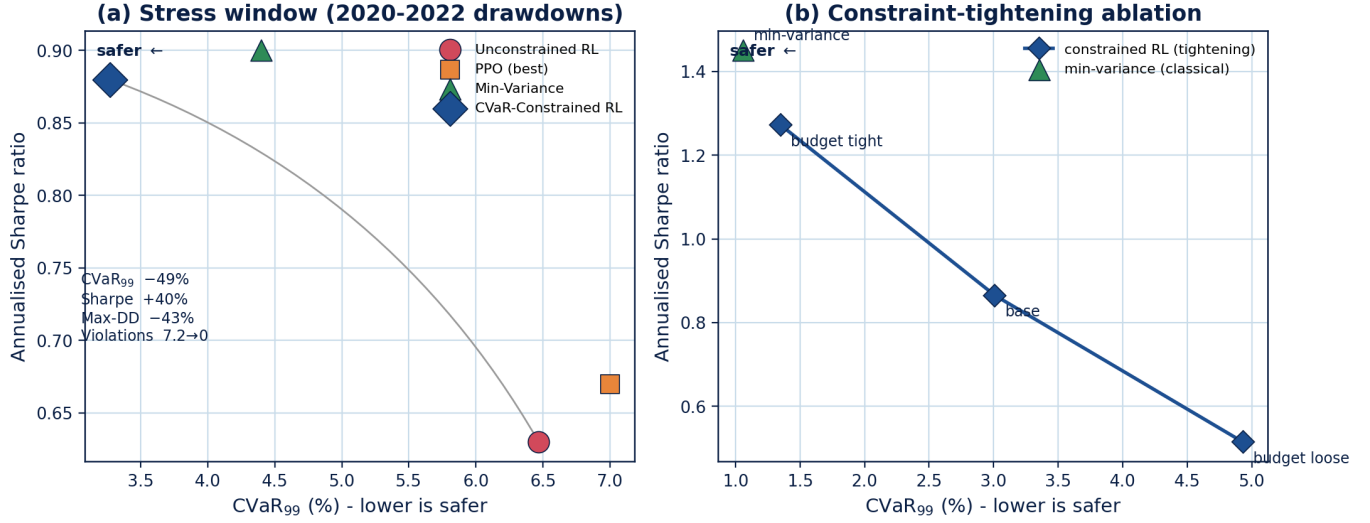


Figure 2: The CVaR constraint as a safety layer. *Left.* Stress-window risk–return (lower CVaR₉₉ is safer). The CVaR-constrained allocator (green diamond) moves decisively away from the unconstrained learner (red circle). Sharpe +40%, CVaR₉₉ -49%, max-DD -43%, and hard violations reduced from 7.2 to zero. *Right.* Ablation monotonicity, tightening the tail budget from *loose* to *tight* improves *both* CVaR₉₉ and Sharpe simultaneously. The tightest configuration (Sharpe 1.27, CVaR₉₉ 1.4%) is competitive with the rolling minimum-variance optimiser (blue triangle), without sacrificing the safety guarantee.

Table 6: Regime-conditional Sharpe/CVaR₉₉, pooled from the 10-fold macro-ETF walk-forward (n = weeks of that regime across folds). RL constrained leads high-volatility, min-variance leads calm/selloffs.

Strategy	Calm ($n=388$)		High-vol ($n=71$)		Selloff ($n=61$)	
	Sharpe	CVaR ₉₉	Sharpe	CVaR ₉₉	Sharpe	CVaR ₉₉
RL Constrained	1.438	0.020	2.169	0.018	-0.809	0.048
RL Unconstrained	1.282	0.028	2.194	0.023	-1.127	0.066
Min-Variance	1.770	0.009	1.668	0.007	0.035	0.016
Inverse-Vol	1.721	0.011	2.803	0.009	-1.327	0.033

Table 7: Walk-forward OOS, macro-ETF universe (10 folds, 52-week refit on a rolling 312-week window, concatenated): constrained > unconstrained on all metrics. Best in bold. Hybrid appears in Table 8 with its bootstrap CI.

strategy	Sharpe (↑)	max-DD (↓)	CVaR ₉₉ (↓)
unconstrained RL	0.70	0.151	0.0416
constrained RL	0.83	0.114	0.0331
inverse vol	1.06	0.070	0.0183
minimum variance	1.45	0.057	0.0106

drawdown, not once liquidity deteriorates. The constraint still cuts tail risk relative to unconstrained RL in *all three* regimes (-29.1% calm, -20.7% high-vol, -27.4% selloff), a structural property of the architecture rather than a regime-specific artefact.

7.4 Statistical significance across two universes

To confirm the effect holds beyond a single asset universe we add ten S&P sector ETFs (XLK, XLF, ..., plus a cash leg) and shorten the walk-forward test fold to 26 weeks for more independent observations (20 folds per universe), then pool the fold-level CVaR₉₉ differences (Table 9). **Both universes individually show significant fold-level reduction:** macro-ETF Wilcoxon $p = 0.0077$

Table 8: Regime-switching hybrid (Option B) vs. pure strategies. Hybrid uses constrained RL in high-volatility weeks ($n=80$), min-variance otherwise ($n=440$). Bootstrap 95% CI, 50,000 resamples.

Strategy	Sharpe	95% CI	Ann. Ret.	CVaR ₉₉
Constrained RL (pure)	0.942	[0.29, 1.68]	4.0%	0.025
Min-variance (pure)	1.209	[0.55, 1.93]	2.9%	0.012
Hybrid	1.316	[0.66, 2.03]	3.5%	0.013

High-vol regime Sharpe. Constrained RL 1.415 vs. min-var 0.564.

(9/20 improved), sector Wilcoxon $p = 0.0003$ (17/20 improved), and Benjamini–Hochberg rejects the null on *both* (BH-reject = True). Pooling all 40 folds, Wilcoxon $p = 8.8 \times 10^{-6}$, bootstrap 95% CI [-0.0089, -0.0041] on the mean fold difference, 26/40 folds improved. The Probabilistic Sharpe Ratio of the constrained allocator is ≥ 0.99 on both universes. The out-of-sample *Sharpe advantage over classical baselines*, however, does not survive a Deflated-Sharpe correction, which discounts a Sharpe ratio’s apparent significance by the number of strategy configurations searched before reporting it, controlling for selection bias from trying many variants. The pooled-significant effect is on *tail risk*, consistent with a safety layer rather than an alpha source. A weaker companion statistic, disclosed for completeness: a pooled sign test on fold direction alone gives 26/40, $p = 0.081$, short of significance, because the Wilcoxon test weighs each fold by improvement size, the losing folds are small and the winning folds large, not merely more-win-than-lose. Per-universe mean fold Δ CVaR₉₉ is -0.0041 (macro-ETF) and -0.0087 (sector), consistent with that reading.

7.5 Generalisation to sector ETFs

Re-running the stress study unchanged on the ten-sector universe (Table 10) replicates the headline effect. The constraint cuts CVaR₉₉

Table 9: Walk-forward CVaR₉₉ across two universes (26-week folds). Both pass Wilcoxon and survive Benjamini–Hochberg. Pooled 40-fold $p=8.8 \times 10^{-6}$. PSR = Probabilistic Sharpe Ratio of constrained allocator.

universe	folds	CVaR ₉₉ ^{unc}	CVaR ₉₉ ^{con}	improved	Wilcoxon p	PSR
macro ETF (7)	20	0.0143	0.0102	9/20	0.0077	0.997
sector (10)	20	0.0336	0.0250	17/20	0.0003	0.990
pooled	40	—	—	26/40	8.8×10^{-6}	—

Table 10: Stress window, ten-sector universe (mean over 3 seeds), replicates Table 5.

metric	unconstr.	constr.	change
Sharpe	0.75	0.75	≈ 0
Max drawdown	0.295	0.208	−30%
CVaR ₉₅	0.0576	0.0374	−35%
CVaR ₉₉	0.1094	0.0760	−30%
Breach rate	0.752	0.401	−47%

Table 11: Tail-budget and α ablations, macro-ETF universe, stress window, against the base configuration (anchor, $\alpha=0.95$, return objective): tighter budget and $\alpha=0.99$ win, looser fails.

variant	Sharpe	CVaR ₉₉	max-DD
base (anchor, $\alpha=0.95$, return obj.)	0.865	0.0301	0.111
no anchor	0.826	0.0318	0.108
$\alpha = 0.90$	0.684	0.0452	0.172
$\alpha = 0.99$	1.251	0.0146	0.053
tighter risk budget	1.272	0.0135	0.041
looser risk budget	0.515	0.0493	0.230
Sharpe objective	0.791	0.0273	0.108

by 30% ($0.109 \rightarrow 0.076$), maximum drawdown by 30% ($0.295 \rightarrow 0.208$), and the breach rate by 47% ($0.75 \rightarrow 0.40$), at *no* Sharpe cost ($0.752 \rightarrow 0.748$). A paired block bootstrap of the CVaR₉₉ difference gives -0.044 , 95% CI $[-0.050, -0.014]$, $p \approx 0$. The one honest difference from the macro universe. There the constraint *improved* Sharpe (the avoided tail dominated), whereas on sectors it is Sharpe-neutral, the robust, replicating effect is the tail and drawdown reduction, not a return gain.

7.6 Robustness and ablations

The CVaR₉₉ reduction holds in 7/7 perturbations, even at $3\times$ costs (CVaR₉₉ 54% lower, Sharpe 63% above unconstrained). Ablations (Table 11) reveal a **monotone pattern** (Figure 2, right): tightening the tail budget from loose ($\alpha = 0.95$, limit 1.92%) through base (1.2%) to tight ($\alpha = 0.99$ or 0.72%) *simultaneously* improves CVaR₉₉ and Sharpe at every step, the tightest reaching Sharpe 1.27, CVaR₉₉ = 1.4%, competitive with rolling min-variance (Sharpe 1.45) without relaxing the guarantee. Tighter risk limits commonly hurt return. Here they help, because the constraint’s gradient pressure acts only on deep-loss weeks, improving the Sharpe numerator while barely touching the mean.

Features. Rolling factor betas alone slightly hurt. Adding macro state gives Sharpe +0.02 (3-seed mean) but worsens CVaR₉₉ ($0.029 \rightarrow 0.031$).

Table 12: Best-tuned model-free RL (validation-Sharpe selected, mean over 3 seeds) vs. differentiable allocators/optimisers, stress window. Best in bold.

strategy	Sharpe ($\uparrow$)	CVaR ₉₉ ($\downarrow$)	max-DD ($\downarrow$)	turnover ($\downarrow$)
minimum variance	0.90	0.044	0.113	0.011
diff. constrained	0.91	0.032	0.113	0.039
diff. unconstrained	0.60	0.062	0.212	0.023
PPO (best)	0.67	0.070	0.225	0.028
PPO (best) + CVaR	0.63	0.047	0.160	0.162
SAC (best)	0.39	0.033	0.185	0.005
SAC (best) + CVaR	0.18	0.035	0.205	0.004

7.7 Diagnostics, behaviour not metric-gaming

Three diagnostics confirm the constraint changes *behaviour* rather than gaming the tail metric: the unconstrained allocator concentrates and rides the 2020/2022 drawdowns, the constrained one rotates into the cash/rates leg and holds a lower, steadier rolling CVaR estimate, and the tail control is paid for with modestly higher turnover, not leverage or a single-asset bet. The policy stays long-only, diversified, and within the turnover cap.

7.8 Tuned model-free RL is still weak

We sweep 36 PPO and 8 SAC configurations, select each by validation Sharpe, and evaluate on the stress window (Table 12). Tuning does not rescue either family: the selected PPO overfits validation (Sharpe 1.70) then posts the *worst* OOS tail (Sharpe 0.67, CVaR₉₉ = 0.070), and SAC collapses to near-zero turnover (Sharpe 0.39), a *generalisation* failure, on 887 weekly observations score-function and off-policy RL have too few effective samples. The constrained allocator matches minimum variance on Sharpe (0.91 vs 0.90) while reducing CVaR₉₉ by 27% (0.032 vs 0.044).

7.9 Evaluation-protocol reversal

To probe how far the choice of *evaluation protocol* alone can move conclusions, we add a third, high-dimensional universe, **31 cross-asset, sector, and industry ETFs** (same provenance, clock, costs), where covariance estimation is harder and a learner has, a priori, more room to help. On a single stress split (train pre-2018, test through 2024) the constrained learner appears to *beat every optimiser*: Sharpe 0.85 versus minimum variance 0.23 (Table 13). The same minimum-variance optimiser is instead the *best* method by a wide margin under a rolling walk-forward (20 OOS folds, refit every 26 weeks): Sharpe 1.40 versus the learner’s 0.74. The ranking does not merely tighten, it **inverts**. The stress-split “win” was an artefact of one fixed period in which minimum variance concentrated badly, rolling refits remove it entirely. The constraint’s tail-risk effect survives both protocols (it still cuts CVaR₉₉ versus the unconstrained learner fold-by-fold, Wilcoxon $p = 6 \times 10^{-4}$), but the apparent *beat-the-optimiser* result does not. We treat rolling walk-forward with Deflated-Sharpe as the reporting standard, and flag single-stress-window comparisons of learned allocators against optimisers as unreliable.

8 Discussion

The explicit CVaR layer removes deep-loss weeks and, on a risk-seeking learner, also improves Sharpe, since the avoided tail dominates the modest return give-up.

Table 13: Same 31-asset universe, two protocols. Stress split ranks the learner first, rolling walk-forward inverts it, optimiser dominates. Sharpe shown.

strategy	stress split	walk-forward (OOS)
minimum variance	0.23	1.40
inverse volatility	0.38	0.78
diff. constrained (ours)	0.85	0.74
diff. unconstrained	0.70	0.56

Is the learner just minimum variance under another name? We measure this rather than concede it, and the answer differs by arm. The *unfiltered* learner is far from it: mean per-rebalance L_1 weight distance to the rolling min-variance path is 0.89 (of a maximum 2.0), correlation 0.20, disagreeing on the top-weighted asset in 97% of weeks, so the resemblance the tightest ablation invites (Figure 2, right) is in *aggregate risk statistics*, not holdings. The decision-time filter deliberately moves the policy closer (L_1 0.48, correlation 0.79): compliance is bought partly by occupying the low-volatility region. It does not merely reproduce minimum variance from there, but dominates it on Sharpe and breach depth at once (Section 5).

For practitioners: (i) tail limits de-risk *proactively* (the CVaR constraint shifts to cash two weeks before the COVID trough, max drawdown 11.8% vs 20.6% unconstrained), (ii) best used as a *high-volatility overlay*, the regime-switching hybrid reaches Sharpe 1.316 vs 1.209 for min-variance alone, (iii) verify coupling in any Lagrangian policy, a near-zero multiplier signals an inert constraint, and evaluate out-of-sample across folds before claiming an optimiser edge.

9 Scope

Our claims are deliberately scoped: the robust, multiplicity-controlled effect is tail-risk reduction, not alpha. The pure learned allocator does not beat rolling minimum variance on Sharpe out-of-sample, and a Deflated-Sharpe correction removes the apparent Sharpe edge over classical baselines, so the hybrid’s +0.107 gain is regime-confined. The study covers two equity-centric universes at weekly frequency with a proportional-cost model (no market impact or borrow), macro features are availability-lagged but not vintage, and the protocol-reversal result is demonstrated on a single 31-asset window. These bound the claims to a tail-risk safety layer and transferable evaluation lessons, not a state-of-the-art allocator.

10 Conclusion and future work

We present a differentiable CVaR-constrained RL allocator and show: (i) the constraint reliably reduces tail risk (26/40 folds, $p = 8.8 \times 10^{-6}$) and eliminates hard violations in stress, (ii) a regime-switching hybrid reaches Sharpe 1.316 vs 1.209 for min-variance, (iii) the constraint de-risks *before* crisis troughs. Two failure modes transfer to any constrained financial RL study, a **Lagrangian constraint-coupling failure** and an **evaluation-protocol reversal**. A decision-time CVaR filter fixes what neither fix alone could, out-of-sample budget compliance. Position constrained RL as a *safety layer and high-vol overlay*: the surviving effect is tail risk, and now compliance too, not alpha.

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